ME2 Computing- Coursework summary

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A) What physics are you trying to model and analyse? (Describe clearly, in words, what physical phenomenon you wish to analyse)

We aim to model the propagation of sound waves (mechanical waves) in a confined space, specifically the instantaneous single-pitch sound generated from a point source, such as Nicolas' voice at the start of lectures. This sound wave travels through the air, reflects off the walls of CAGB 203, and interacts with the environment. By utilizing the 2D wave equation, describing sound wave movement in air within a two-dimensional space, we can analyse wave behaviours, including reflection and dispersion, aiding our understanding of acoustic phenomena like echoes and sound energy distribution. Assuming the lecture room is square, and Nicolas' voice is a brief "note" (single-pitch pint source), we apply Dirichlet boundary conditions at the walls to model the phenomenon. The initial sound wave is modelled by the function *f(x,y)*, determining its amplitude, position, and shape.

B) What PDE are you trying to solve, associated with the Physics described in A? (write the PDE)

2D wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) \tag{1}$$

This is a hyperbolic PDE in 2D. 2nd order in space, and 2nd order in time.

C) Boundary value and/or initial values for my specific problem: (be CONSISTENT with what you wrote in A)

The following conditions are set:

$$u_{[0,x,y]} = f(x,y) = e^{(-350*((x-1)^2+(y-1)^2)} \tag{2}$$

The initial condition (equation 2) describes the wave shape at t=0, i.e. the distribution of sound in the room at the immediate moment that Nicolas starts speaking. Such Gaussian model is often used to model sound attenuation phenomena; it peaks at the centre and smoothly decays towards zero, in line with the starting scenario; sound emitted from a localised source that propagates in the room. The source is at Nicolas' position in the lecture room [0.5 , 0.25].

$$\left.\frac{\partial u}{\partial t}\right|_{t=0} \approx 0 \tag{3}$$

Equation 3 is the initial velocity of the wave (Neumann condition). When someone speaks, the initial velocity is technically non-zero. However, its magnitude is many orders of magnitude smaller than the speed of sound through air. In room acoustics and speech modelling, where the emphasis is on lower-frequency sound waves and long-time scales, the initial velocity of sound is typically negligible compared to the speed of sound in air. Therefore, approximating the initial velocity to zero is a simplification that balances computational efficiency with accuracy of application.

$$u_{[0,y,t]} = 0 \tag{4}$$

$$u_{[1,y,t]} = 0 \tag{5}$$

$$u_{[x,0,t]} = 0 \tag{6}$$

$$u_{[x,1,t]} = 0 \tag{7}$$

The remaining boundary conditions described by equations 4 to 7 show zero displacement at the boundaries/walls. This follows from the assumption that the walls are perfectly reflecting surfaces: they don't transmit nor absorb wave energy: the incident wave is fully and perfectly reflected, without attenuation or phase change. This is a reasonable assumption, especially in the context of the lecture hall where echo is the most significant phenomenon (especially if empty). It's worth noting that the derivative at the walls is non-zero.

D) What numerical method are you going to deploy and why? (Describe, in words, which method you intend to apply and why you have chosen it as opposed to other alternatives)

The equations are solved numerically by employing central finite differences on the partial derivatives in Equation 1. This approach establishes a relationship between a point's value and the values of its surrounding points that affect it. Subsequently, the point to be determined in "the future" is isolated from the points at previous time steps.
After evaluating the primary characteristics of both methods, an explicit approach has been preferred over implicit.

Table 1 : pros and cons of explicit vs implicit

EXPLICIT	**IMPLICIT**
Algorithm less complex	More complex algorithm
Convergence and stability aren't guaranteed	Convergence is unconditional
Not suitable for stiff problems (time step required would be very small)	The method is 1st order accurate and loses accuracy when time step increases

As shown in Table 1, the primary advantage of the implicit method over the explicit one is its unconditional convergence. However, this comes with a trade-off in computational efficiency. Implicit methods employ more complex algorithms and demand greater computational power. If an appropriate convergence criterion is met, explicit methods may prove more advantageous. In our case, where the problem isn't 'stiff,' meaning it doesn't demand very small and computationally expensive time steps for convergence, the explicit method is adequate.

E) I am going to discretise my PDE as the following (show the steps from continuous to discrete equation and boundary/initial conditions):

We start from the continuous 2D wave equation (equation 1, recalled here):

$$\frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) \tag{1}$$

Where u is:

- A scalar function of time and space $u(t, x, y)$
- Displacement of the wave from its equilibrium position (in the case of a sound wave, it represents the variation in air pressure caused by the wave)

x and y are the position in space, t is time, and c is the wave speed (in air ≈343m/s).

Firstly, we introduce a mesh in time and space.
In time it consists of time points:

$$t = 0 < t_1 < \cdots < t_N \tag{8}$$

with constant time spacing $\Delta t = t_{n+1} - t_n = k$.
And in space:

$$x_o < x_1 < \cdots < x_{Nx} \quad \text{(x direction)} \tag{9}$$

$$y_o < y_1 < \cdots < y_{Ny} \quad \text{(y direction)} \tag{10}$$

where a constant mesh size is used: $\Delta x = x_{i+1} - x_i$. And applied to a square domain: $\Delta x = \Delta y = h$.
The unknown u at a mesh point t_n, x_i, y_j is denoted by $u_{i,j}^n$.

Before discretization, the equation is non-dimensionalised to simplify the problem, aid in scaling analysis, identify dimensionless parameters (if any), and reduce computational effort, enhancing result interpretation. The dimensionless form of the wave equation is as follows:

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \tag{11}$$

Where $0 < x, y < 1$

E) E) I am going to discretise my PDE as the following (show the steps from continuous to discrete equation and boundary/initial conditions) (Contd…):

The difference quotients corresponding to the partial derivatives are approximated using central finite difference in time and space (equations 3 and 4):

$$\frac{\partial^2 u}{\partial t^2} \approx \frac{1}{k^2}[u\,(t+k,x,y) - 2u(t,x,y) + u(t-k,x,y)] \tag{12}$$

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{1}{h^2}[u\,(t,x+h,y) - 2u(t,x,y) + u(t\,,x-h,y)] \tag{13}$$

$$\frac{\partial^2 u}{\partial y^2} \approx \frac{1}{h^2}[u\,(t,x,y+h) - 2u(t,x,y) + u(t\,,x,y-h)] \tag{14}$$

Central difference was preferred because it is stable, accurate, and fast converging.
Stability is of importance since an explicit method is employed, hence central difference is more adequate.

Substituting equations 12 to 14 into equation 2 and writing in index notation leads to equation 15:

$$\frac{1}{k^2}[\,u_{i,j}^{n+1} - 2u_{i,j}^{n} + u_{i,j}^{n-1}\,] = \frac{1}{h^2}[u_{i+1,j}^{n} + u_{i-1,j}^{n} + \,u_{i,j+1}^{n} + u_{i,j-1}^{n} - 4u_{i,j}^{n}] \tag{15}$$

Letting $r = \frac{k^2}{h^2}$ and rearranging equation 15 becomes:

$$[\,u_{i,j}^{n+1} + u_{i,j}^{n-1}\,] = \,r\big[u_{i+1,j}^{n} + u_{i-1,j}^{n} + \,u_{i,j+1}^{n} + u_{i,j-1}^{n}\big] + (-4r+2)u_{i,j}^{n} \tag{16}$$

This equation is insightful when determining the condition for stability; since the contribution of the $u_{i,j}^{n}$ term must be positive, $-4r + 2 \geq 0$ must hold, hence $r \geq \frac{1}{2}$. At limit r = $\frac{1}{2}$, the final discretised equation becomes:

$$u_{i,j}^{n+1} = \frac{1}{2}\big(\,u_{i+1,j}^{n} + u_{i-1,j}^{n} + \,u_{i,j+1}^{n} + u_{i,j-1}^{n}\big) - \,u_{i,j}^{n-1} \tag{17}$$

For the initial condition, an issue arises with the final $u_{i,j}^{n-1}$ term. This value lies outside the mesh at negative time $t = -k$; such ‘ghost value’ can be determined from the central finite difference of the Neumann BC in Equation 3:

$$\frac{\partial u}{\partial \mathrm{t}} \approx \frac{1}{2k}\big(\,u_{i,j}^{n+1} - u_{i,j}^{n-1}\big) = 0 \tag{18}$$

which gives that $u_{i,j}^{n-1} = u_{i,j}^{n+1}$. A special case (equation 19) is derived for t = 0:

$$u_{i,j}^{1} = \frac{1}{4}(u_{i+1,j}^{0} + u_{i-1,j}^{0} + \,u_{i,j+1}^{0} + u_{i,j-1}^{0}\,) \tag{19}$$

Equation 17 can be used for the t > 0 terms. The scenario is fully defined: all the terms at t = 0 can be determined from the initial and boundary conditions, and it’s possible to iterate up until the desired time duration.

F) Plot the numerical results comprehensively and discuss them (discuss how the results describe the physics and comment on any discrepancies or unexpected behaviours). Use multiple types of visual graphs. Present and discuss any outcomes of the grid analysis, as requested in Task 9, too.

The wave develops in the room as per [this animation](). Snapshots of the wave propagation at different times are shown in Figures 1 to 4. This is also plotted as [a contour animation]() developing over time.

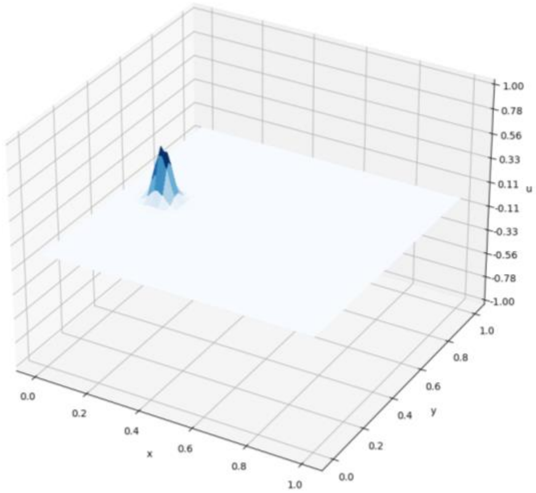

Figure 1: acoustic wave source at t=0s

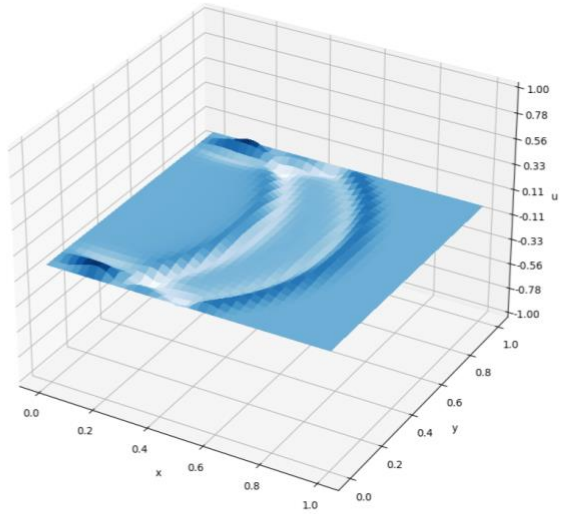

Figure 2: acoustic wave at t = 0.06s

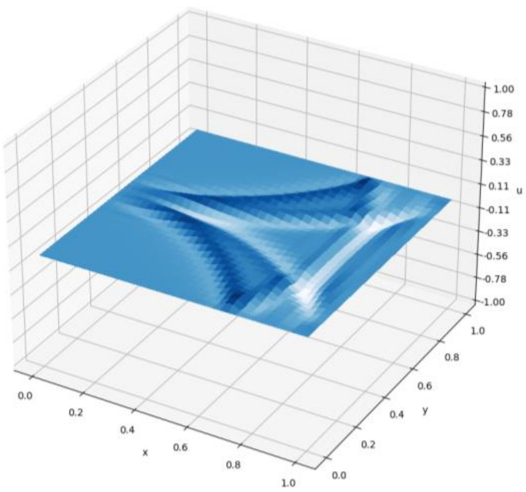

Figure 3: acoustic wave at t=0.1s

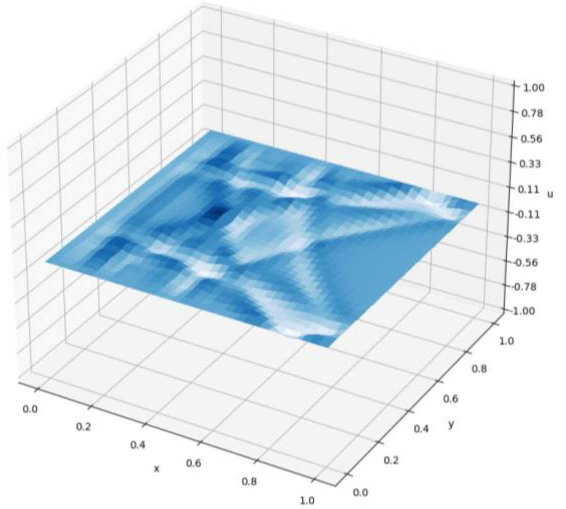

Figure 4: acoustic wave at t=0.18s

The source contour and two developement contours are also presented in Figure 5 to 7 (page 5).

F) Plot the numerical results comprehensively and discuss them (discuss how the results describe the physics and comment on any discrepancies or unexpected behaviours). Use multiple types of visual graphs. Present and discuss any outcomes of the grid analysis, as requested in Task 9, too. (Contd…)

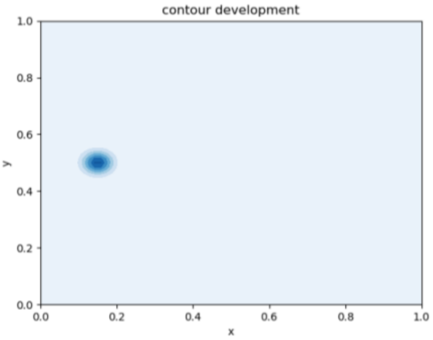

Figure 5: contour source (t=0s)

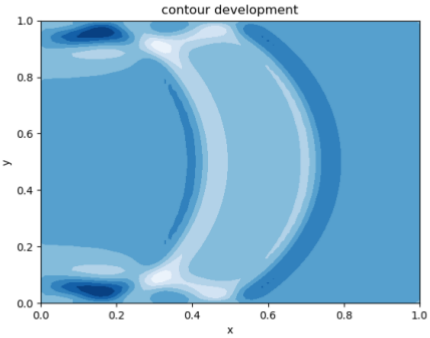

Figure 6 : contour at t=0.06s

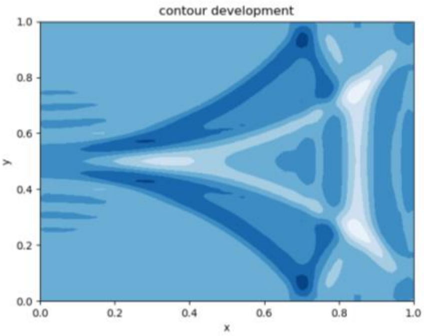

Figure 7 :contour at t=0.1s

Understanding the wave's impact on a specific plane is valuable and can be achieved by "slicing" the space at a particular point. In this case, we chose the vertical plane parallel to the x-axis, passing through the source point. Figures 8 to 10 depict three moments along this plane.

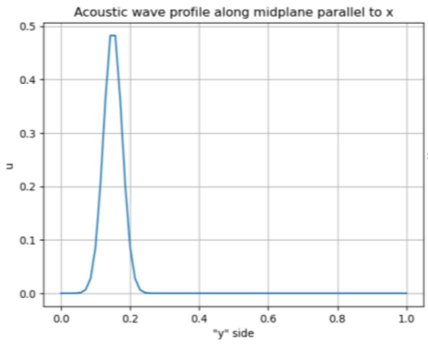

Figure 8 : plot at t=0s

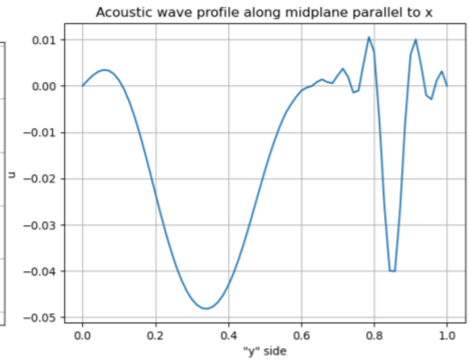

Figure 9: plot at t=0.1s

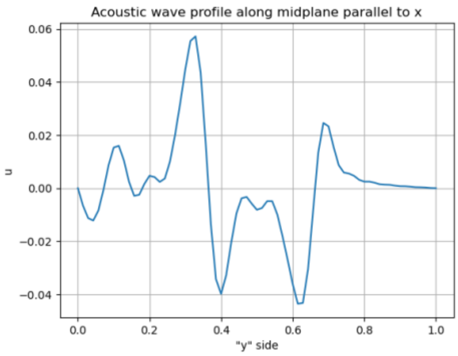

Figure 10: plot at t=0.18s

The plots reflect the physics accurately: Figure 1 and 8 depict a wave modelled as an instantaneous emission with a single peak at the source position at time = 0, assuming quiet surroundings. Subsequently, the wave reflects from the walls, generating new profiles. According to the principle of superposition, waves may interfere constructively (combining crests of the same sign) or destructively (subtracting opposite-signed crests), resulting in amplified peaks or neutralized areas. Figures 3 and 7 (both plot and contour) exemplify these phenomena: the darker, symmetrical area around the source illustrates constructive interference, while the lighter/white area demonstrates destructive interference, neutralizing peaks.
The wave propagation, hence, as expected.

F) Plot the numerical results comprehensively and discuss them (discuss how the results describe the physics and comment on any discrepancies or unexpected behaviours). Use multiple types of visual graphs. Present and discuss any outcomes of the grid analysis, as requested in Task 9, too. (Contd…)

For first grid size:

- $h = 0.01429$ (70 intervals)
- Convergence criterion: $k_{max} = 0.0101$
- $k = 0.001$

The time elapsed to compute the numerical method is T = 8.91 s.

For a 10x mesh size reduction the effect of a finer mesh yields to smoother surfaces/curves generated (Figure 11). This fineness however leads to an increased computational time by a factor of 130x, rendering the computation too slow. In addition, k_{max} =0.00101 for convergence, whilst the time step in use here is k=0.001; this lies very close to the limit of convergence. This grid size hence highlights a limitation of the explicit method that will be discussed later in Section G.

For an even greater reduction of 50x, k_{max}=0.0002. If the utilised time step remains 0.001, the method does not converge; this case is again further discussed in Section G.

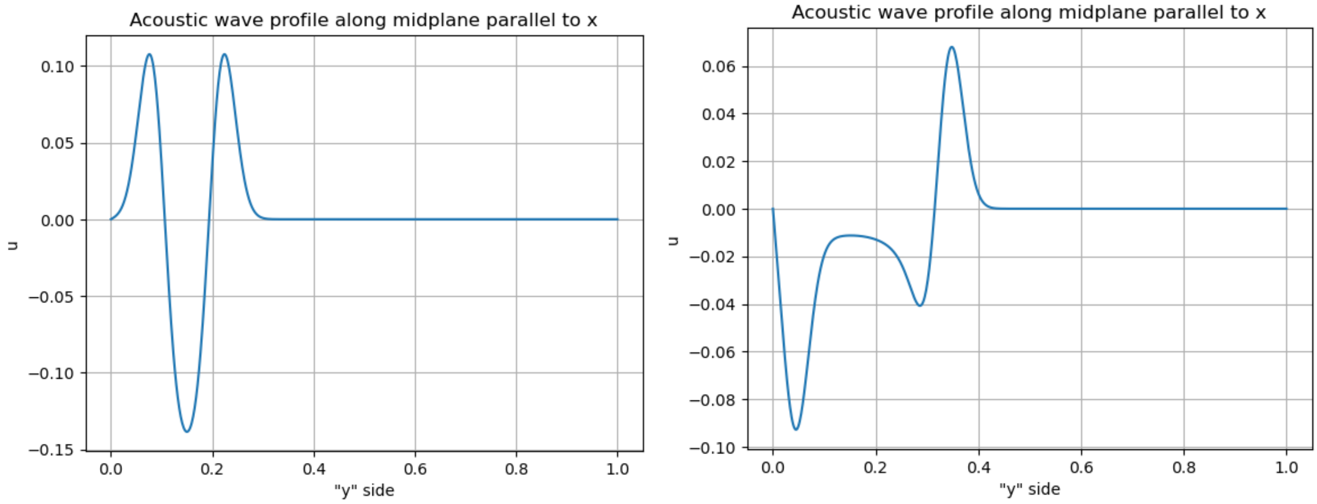

Figure 11: wave generated with 10x mesh size reduction in space at different time snapshots.

G) Other remarks (limits of the model, convergence problems, possible alternative approaches, anything you find relevant and important to mention):

The model performs better with a relatively large mesh size. When the spatial grid size is reduced (for example the 10x reduction in Section F), the computational time increases exponentially. This highlight time as primary constraint within the convergence regime when reducing mesh size.

The other main limitation was already determined initially: the time step is bounded by the convergence criterion – defined during the method evaluation – to a specific value depending on the mesh size. This means that a relatively small mesh size requires a very small time step to account and mitigate the convergence problem. However, approaching the divergence limit for the fixed time step chosen of 0.001 s, the grid and time step are small enough to shift the primary concern from convergence to computational time.